Modelling the combined effect of moisture and temperature on secondary infection in a coupled host-pathogen FSPM

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***Abstract*—Weather conditions are an important driver of disease development. For example for yellow spot in wheat, warm and moist conditions favour secondary infection. Although the relationship between environment and disease development is the basis of many epidemiological models, changes in plant architecture and growth have an effect on disease progress and severity as well. Functional-structural plant models (FSPMs) are well suited to study the interactions between pathogen, climatic conditions and growing host crop. In this study we focused on simulating the effect of weather conditions on the progression of secondary infection in yellow spot and the interaction with growing wheat canopy. Simulations were performed using a coupled host-pathogen FSPM with standard meteorological data input. The model develops on previous coupled host-pathogen FSPMs by combining response functions to temperature and wetness duration and calculating the hourly progression of secondary infection. The simulated diseased area differed with different combinations of temperature and moisture response models. Changes in dispersal pattern were observed mainly in relation to spore release rate.**

***Index Terms*—moisture, leaf wetness duration, temperature, yellow spot, tan spot, disease modelling, functional-structural plant modelling.**

I. Introduction

Functional-structural plant models (FSPMs) integrated in host-pathogen models allow us to study and test hypotheses about interactions between disease development, plant architecture and growth, and environment (Fig. 1).

Recently, Robert *et al.* [1] used an FSPM approach to study effects of traits related to plant architecture on disease progression of Septoria tritici blotch in wheat, spread by rain-splash. Their model could reproduce, e.g., negative effect of internode length and of stem elongation rate, or positive effect of leaf length, on disease progress on upper leaves. They also found that the impact of plant traits on epidemics depended on weather conditions. Some architectural traits had lower impact when weather conditions were unfavourable for disease. Timing of plant development (leaf emergence, leaf senescence) and timing of rain events were important for dispersal and further pathogen development.

If climatic conditions are favourable, the secondary infection cycle can repeat several times in a lifetime of the disease and result in rapid development of the disease in the growing crop [2]. Environmental variables and their combined effect were used as a driver in many plant disease prediction models [3]–[8]. In some examples of coupled host-pathogen FSPMs, infection was modelled as a function of favourable temperature (combined with leaf age) [9], or moisture (modelled as a combination of high humidity and upper irradiance threshold for a certain period of time [10]), or as a combination of optimal temperature range and moisture (high relative humidity) for a period of time [11]. An exception example were diseases in tropical regions, where weather conditions were not the limiting factor for infection [12].

Yellow spot (tan spot) is an important foliar wheat disease throughout the world and disease development is strongly related to temperature and moisture conditions [13]. In the coupled host-pathogen FSPM for yellow spot in wheat [11], disease response to climatic conditions of different locations of Western Australia, evaluated for a number of consecutive hours within a day, showed to be strongly dependant on timing and interruptions of favourable conditions.

In this study we looked in more detail at interactions between weather and epidemics, and resulting spatially-explicit disease progression over time (during growing season) and space (upwards within a plant, and inside a growing canopy, modelled in 3D). We refer to this as disease progression in 4D. To study the interactions, we modified the existing pathogen model [11] in two ways: 1) use of different response functions to moisture (expressed by leaf wetness duration) and temperature, and their combination; 2) continuous hourly-based evaluation of weather data for the calculation of secondary infection progression (as opposed to an evaluation in discrete steps for each day), including spore loss by wash off and desiccation. We investigated different combinations of temperature and moisture response models, and different spore release rates.

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II. Methods

A. Model overview

We used a coupled host-pathogen FSPM which was developed to study interactions between pathogen, climatic conditions and growing host crop, on an example of yellow spot in wheat (Fig. 1, [11]). The model combined an epidemiological model of *Pyrenophora tritici-repentis*, the fungal pathogen that causes yellow spot, with an FSPM for cereal crops, including a spatially-explicit model of spore dispersal in three dimensions. All parts of the coupled model (epidemiological model, FSPM, dispersal model) were driven by meteorological data.

In the model, plant development is described by rewriting rules and is temperature-driven. Leaf carbon assimilation is calculated as a function of absorbed radiation, computed by a Monte Carlo path tracing method. Plant organ growth is calculated using a beta growth function, based on available assimilates. The basic structural unit, and a link to the pathogen submodel, is a leaf segment, where a leaf is made up of a sequence of rectangular leaf segments. The pathogen submodel starts with a primary inoculum placed on one or more leaf segments of a seedling (at a given day of year or as calculated by the ascospore maturation model [14]). After a succesful infection, secondary cycle can repeat several times through the season if the conditions are favourable. It consists of a latency period, during which lesions (resulting from successful infections) grow linearly as a function of thermal time, followed by a sporulation. Spores are released from the lesions into the canopy and are deposited on visible leaf segments using a cone approach [9], [15]. Deposited spores can be a source of new secondary infection, if relative humidity and air temperature are above a threshold value for a certain period of time.

In this study, time step of the pathogen submodel was modified to be one hour, time step of the plant submodel and of the coupled model is one day. The model was implemented in the modelling platform GroIMP [16]. Details on pathogen response to climatic conditions and evaluation of the progression of secondary infection are given below.

B. Weather data

Hourly weather data were obtained from a weather station of the Department of Primary Industries and Regional Development (DPIRD), in South Perth, Western Australia, for the year 2016. We took the input weather data for the time period from May 23 till October 25, 2016 (late autumn to spring), as an example of a growing season for this site. For the pathogen submodel, values of air temperature, relative humidity, rainfall, dew point temperature and wind direction (at 13:00 h), were used.

C. Effect of temperature

To account for effect of temperature, T, we considered three previously published response models [5], [7], [9]:

1) Response model 1 (T_CAUB): The response to temperature is modelled as a linear interpolation between four cardinal temperatures [5]: minimum, T_{min}, first optimum, T_{opt1}, second optimum, T_{opt2}, and maximum temperature, T_{max} (Table I, Fig. 2).

2) Response model 2 (T_CALO): The effect of temperature is modelled as a function of normalised temperature [9], as follows:

$$f(T_n) = \frac{(M+N)^{(M+N)}}{N^N M^M} T_n^N (1-T_n)^M \tag{1}$$

with

$$T_n = \frac{T(t) - T_{min}}{T_{max} - T_{min}} \tag{2}$$

where $T(t)$ is temperature at given time step t, T_{min} and T_{max} are cardinal temperatures for infection, and M and N are shape parameters of the curve (Table I, Fig. 2).

3) Response model 3 (T_DUTH): Temperature response is described as follows [7]:

$$f(T(t)) = E' \frac{\exp[(T(t)-F)G/(H+1)]}{1+\exp[(T(t)-F)G]} \tag{3}$$

with

$$E' = E[(H+1)/H]H^{1/(H+1)} \tag{4}$$

where $T(t)$ is temperature (at given time step t), and the parameters E, F, G and H characterize the shape of the response curve (Table I, Fig. 2).

D. Effect of leaf wetness duration

Effect of leaf wetness duration was modelled using a Weibull function [5], [7], as follows:

$$f(LWD) = A\big(1 - \exp\{-[B(LWD - C)]^D\}\big) \tag{5}$$

where LWD is leaf wetness duration, and A, B, C and D are parameters of the response curve (Table I, Figure 3).

E. Estimating leaf wetness duration

Different empirical methods were previously proposed with an aim to estimate leaf wetness duration from (easily accessible) meteorological data [17], [18]. We considered three of them, taking hourly weather data as input:

1) Constant RH threshold (RH_CONST): LWD equals the number of consecutive hours with relative humidity, RH, above some constant threshold, e.g. $RH \geq 90\%$ [17] (alternative threshold values were 87% and 71%, the later derived for agricultural grassland [19]).

2) Dew point depression (DPD): $DPD = T - T_d$, where T_d is the dew point temperature. LWD equals the number of consecutive hours with DPD within two limits, $DPD \leq 1.8°C$ (wetness onset) and $DPD \geq 2.2°C$ (wetness offset) [17], [20].

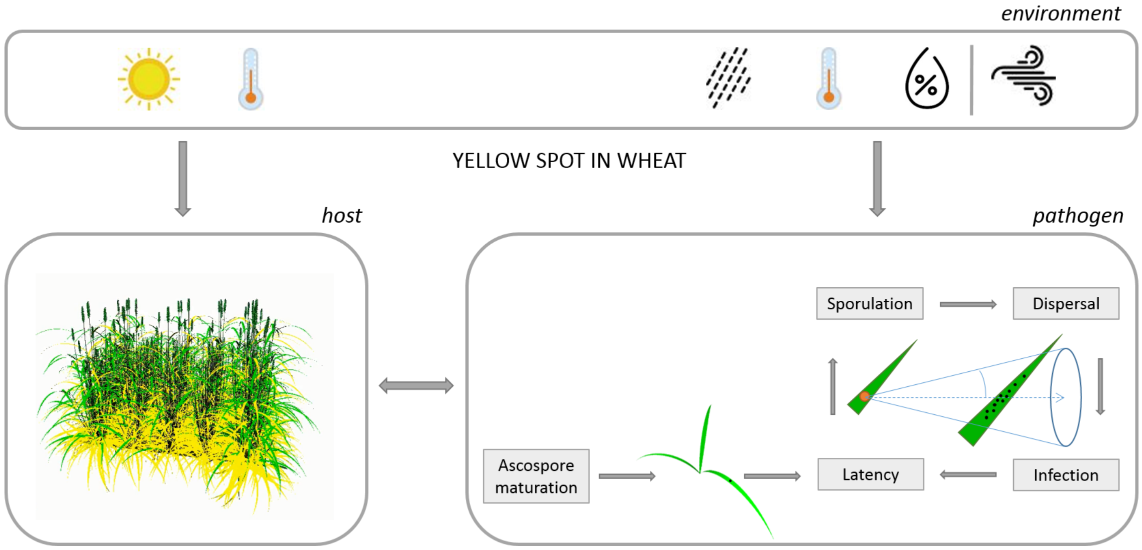


Fig. 1. Schematic overview of the host-pathogen model, driven by environmental input from weather data (daily values are used for the development and growth of the host plant, hourly values for the pathogen's life cycle). Substages of the disease cycle are shown (black circle represents deposited spore, brown circle a lesion after sporulation). Spores released from a lesion after a sporulation event will be deposited on visible leaf segments, i.e. leaf segments lying inside an infinite virtual cone with its tip located at the source lesion, central axis pointing in the wind direction, and characterised by a half opening angle (adapted from [9], [15]).

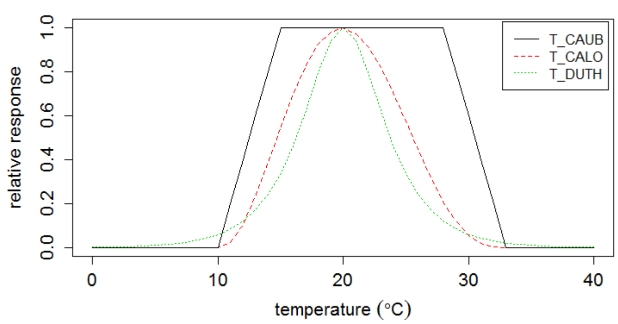


Fig. 2. Relative response to temperature. For the *T_CAUB* model, the range of optimum temperature for secondary infection is between 15 and 28°C. For the *T_CALO* and *T_DUTH* models, the optimum temperature is 20 °C.

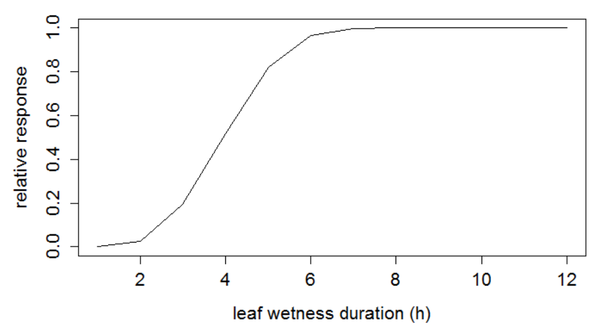


Fig. 3. Relative response to leaf wetness duration. Optimal duration is set to be above 6 hours.

3) Extended RH threshold (RH_EXT): To estimate leaf wetness duration, the base constant value of $RH = 87\%$, above which the leaves were considered wet, is extended by an arbitrary threshold taking into account the rate of change of relative humidity [17], [21]. For values between 70 and 87%, leaves were assumed to be wet if there was an increase in relative humidity by more than 3% in 30 min. Leaves were assumed to be dry if there was a decrease in RH by 2% in 30 minutes. The increase and decrease rate were modified for hourly values, to 6% and 4% per hour, respectively. For RH below 70%, leaves were considered to be dry.

F. Combined effect of temperature and leaf wetness duration

Combined effect of temperature T and leaf wetness duration LWD was computed as the product of them [5]–[7] (Fig. 4), i.e. parameter A in (5) is substituted by $f(T)$:

$$f(T, LWD) = f(T)f(LWD) \tag{6}$$

G. Secondary infection

For the simulations of secondary infection, hourly weather data were evaluated. We used two approaches to estimate if the infection was successful, i.e. discrete and continuous evaluation, executed on daily and hourly basis, respectively.

In Western Australia, secondary infection in yellow spot occurs after favourable conditions of moisture and temperature, with 6 to 12 hours of leaf wetness (or dew) and temperature between 15 and 28°C [22]. For simplification, six hours of suitable conditions were considered as sufficient for infection.

1) Discrete evaluation: After each day if there were six consecutive hours of favourable conditions of moisture and

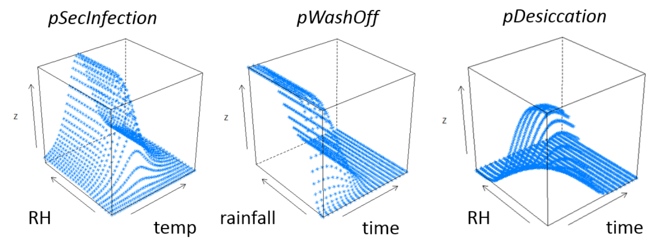


Fig. 4. Infection probability as a function of temperature and leaf wetness duration (expressed using, e.g., relative humidity), dying probability of spores (conidia) by wash off and desiccation (from left to right).

temperature, the day was considered to be favourable for infection. In the model, favourable temperature range lied between 15 and 28°C and five models/parameterisations were used to estimate leaf wetness duration: 1-3) *RH_CONST* with RH of 90%, 87% and 71%, 4) *DPD*, 5) *RH_EXT* with RH above 87% for wet conditions and RH below 70% for dry conditions.

2) Continuous evaluation: Infection progression was simulated using the response function to the combined effect of temperature and moisture, including a calculation of a loss of spores (conidia) due to wash off and desiccation. The procedure was executed after sporulation occurred, as follows.

Calculate every hour:

- Infection rate using a combined response function to temperature and leaf wetness duration (Fig. 4); the value was divided by six, because we assumed that six hours of suitable conditions were needed for a successful infection;
- Loss probability by wash off and by desiccation as a function of cumulated infection factor (infection progression) (Fig. 4, adapted from [8]);
- Survival factor (after loss).

Calculate every day:

- Difference in infection factor between this and previous day;
- Portion of spores (of those released from mother lesion) that will die due to unfavourable conditions, computed from the survival factor;
- Portion of remaining spores that survived unfavourable conditions; from the remaining spores, portion of (penetrating) spores that will cause new infection, based on the infection rate;

When the value of infection progress was equal or greater one, the infection was considered as successful. Spores that survived and caused new infection were the start of the next infection cycle.

III. Results

A. Weather conditions

Fig. 5 shows the input weather data for relative humidity, air temperature, rainfall and wind direction during the growing season.

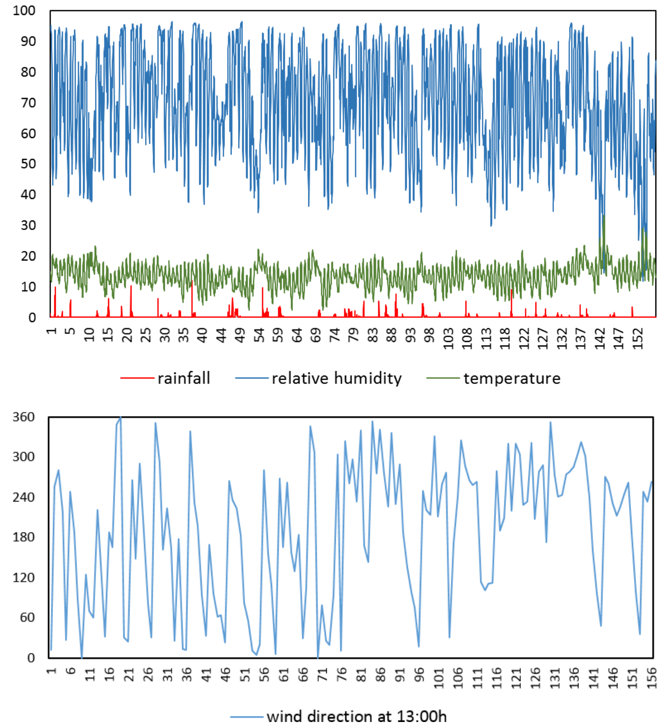


Fig. 5. Weather conditions for the time period between May 23 and October 25, 2016, in South Perth, Western Australia (x-axis shows elapsed days since the start of growing season/simulations): hourly values (above) for relative humidity (in %, blue), temperature (°C, green), rainfall (mm, red); daily values (below) for wind direction at 13:00h (in °).

B. Favourable days for infection

Empirical models for the effect of temperature, leaf wetness duration, their combination, and their effect on secondary infection, were implemented in GroIMP and added to the coupled host-pathogen FSPM. Simulations were run using different model combinations and parameterisations.

1) Discrete evaluation: When we evaluated the weather conditions on daily basis, the number of favourable days increased with decreasing the relative humidity threshold (Table II, Figure 6a), as expected. Results of the *DPD* model were similar to *RH_CONST* with 87%. Results of the *RH_EXT* lied between *RH_CONST* with 87% and 71%. We considered the *RH_EXT* as the default model for later simulations. The results indicate that the favourable conditions were often interrupted and there were only seldom events of six consecutive hours of favourable conditions within a day (in the plot corresponding to the relative value of 1).

2) Continuous evaluation: Fig. 6b shows the evaluation of the combined weather conditions for three different temperature response models, combined with *RH_EXT* (optimal conditions correspond to the relative value of 1). The value of the response factor, divided by 6, corresponds to the hourly increase of infection factor (infection progression).

C. Infection progress (continuous evaluation)

Fig. 7 shows relative values of diseased (lesion) area for combinations of different temperature response models and

TABLE I
LIST OF PARAMETERS FOR THE SECONDARY INFECTION MODEL (REASONABLE VALUES WERE ESTIMATED FOR SOME PARAMETERS).

Definition	Symbol	Value	Adapted from
Temperature (T) effect			
Minimum, maximum T (°C)	T_{min}, T_{max}	10, 33	[5], [9], [23]
First (bottom), second (upper) optimum T (°C)	T_{opt1}, T_{opt2}	15, 28	[5], [22]
Shape parameters	M, N	3.05, 2.3	[9], est.
Max. response at opt. T, degree of asymmetry,	E, H	1, 1	[7]
Optimum T (°C), decline rate,	F, G	20, 0.7	[7], est.
Leaf wetness duration (LWD) effect			
Upper limit, increase rate, delay, inflection point	A, B, C, D	1, 0.3, 0, 3	[5], [7], est.
Conidia loss			
Desiccation effect	B_D, D_D, E_D	4.5, 3.2, 0.05	[8], est.
Wash off effect	B_W, D_W, E_W	7.5, 3.2, 4	[8], est.

TABLE II
NUMBER OF FAVOURABLE DAYS FOR INFECTION DURING THE GROWING SEASON, CALCULATED ON DAILY BASIS, AS A COMBINATION OF 6 CONSECUTIVE HOURS OF SUITABLE TEMPERATURE (BETWEEN 15 AND 28°C) AND LEAF WETNESS, ESTIMATED USING DIFFERENT EMPIRICAL MODELS.

LWD model (threshold)	# Favourable days	Adapted from
RH_CONST ($RH \geq$ 90%)	3	[17]
DPD	5	[17], [20]
RH_CONST ($RH \geq$ 87%)	6	[19]
RH_EXT	9	[17], [21]
RH_CONST ($RH \geq$ 71%)	14	[19]

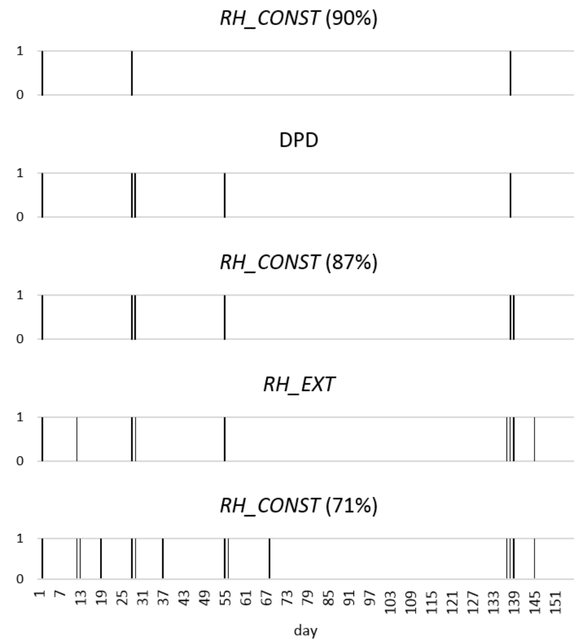
leaf wetness models with different relative humidity thresholds, after they were linked to the crop model. For given climatic conditions, simulated diseased area was highest with the *T_CAUB* model, for any of the tested relative humidity thresholds. Simulated values for the constant threshold of 71% were highest for any of the temperature models, and lowest when constant threshold of 87% or 90% was taken. It was surprising that the simulation results with, e.g., *RH_CONST* of 71% and *RH_EXT*, were lower for *T_CALO* than for *T_DUTH*. This can be explained by looking more closely at climatic conditions after the sporulation and dispersal event on day 42 since the start of simulations, and at the temperature response models as well. Fig. 8 shows the differences in the response models and the resulting (cumulated) infection progress, for the fixed *RH_EXT* combined with three temperature response models. When the temperatures stayed around T_{min}, the response calculated by *T_CALO* was the lowest from the three temperature response models (Fig. 2), and resulted in the lowest increase in the progression. When a rainfall occurs, its effect is higher if the infection is in the low stage of progression (Fig. 4), what was the case for the simulation with *T_CALO*. The combination of the responses to low temperature and to rainfall resulted in high spore loss. *T_DUTH* computed low response to temperatures close to T_{min}. This slowed down the infection progress, but did not stop it.

Increasing the percentage of the spore release rate from 40% to 100% per day resulted in faster disease progress with more diseased area (for *RH_EXT*). However, the disease pattern of the later was slightly less spread towards the canopy borders (Fig. 9), what could be explained by setting the direction of the wind only once, i.e. all spores would be released and dispersed in one event after the sporulation, taking the value for wind direction on that day (Fig. 5).

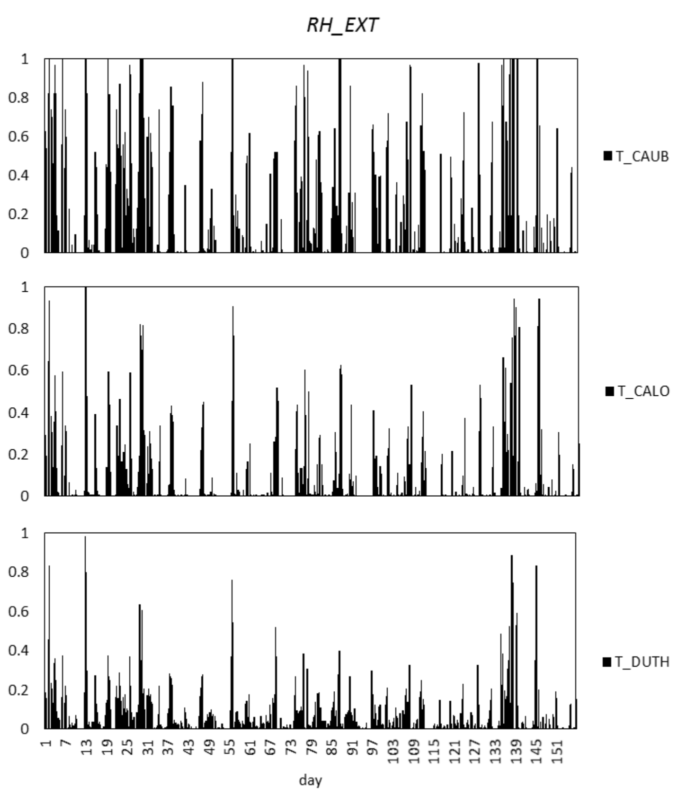
IV. DISCUSSION

We presented an extension of the pathogen submodel from a coupled host-pathogen FSPM to enable refined simulation of secondary infection progression in a 3D crop canopy, with a time step of one hour. A drawback of the earlier model version, which used the discrete evaluation of favourable conditions, was that it ignored the continuous manner of climatic conditions. As a basis for the model extension, we used a disease response model to the combined effect of two weather factors, temperature and leaf wetness duration [7], and considered several published empirical models for temperature response and for estimating leaf wetness duration. Simulation examples showed that stacking models together can result in a strong combined effect (for given parameterisation), e.g., canopy could escape the disease if a temperature response model has zero response to the lower range of temperatures, followed by higher wash off probability in the early stage of infection (Fig. 8, Fig. 9, *T_CALO*). Even without modifying the architectural traits in the coupled model, we could observe differences in disease progression in terms of total diseased area (by the use of different weather response models) or dispersal pattern (because of the more frequent input for wind direction with lower release rate).

Future work will include model validation and further model improvement. Performance of empirical models for estimating leaf wetness duration was reported to vary with site location [17], [25]. Experiments were performed to observe how yellow spot disease progresses from an infected middle plant through a field plot over a growing season (in South Perth). Data analysis from the field experiments is underway and will be used to choose the best response model and to evaluate



(a) Discrete evaluation



(b) Continuous evaluation

Fig. 6. Discrete and continuous evaluation of conditions favourable for infection during the growing period (black bars), using different temperature response models. Note, that the temperature range for the discrete evaluation is the same as the optimum temperature range in *T_CAUB*. However, *T_CAUB* takes into account temperatures lying left and right from the optimum range, as described in the Methods section. In the continuous evaluation, the rate of the response was plotted after each simulation day.

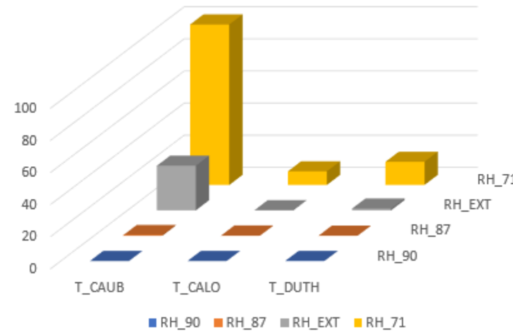


Fig. 7. Relative values of diseased area (after 80 days of simulations, for a test plot with 6x60 plants and spore release 40% per day [24]) for different combinations of relative humidity thresholds and temperature response models.

simulated model outputs. It could also help to evaluate if the model can benefit from a more realistic wind dispersal model, which considers upward movement close to a sporulating source lesion, turbulent fluctuations or even a computation of spore trajectories.

The model takes meteorological data measured above the canopy as input for the disease model. Calculating microclimate inside the 3D canopy could provide better estimate of climatic conditions to which the pathogen is exposed. It would be interesting to include the effects of architectural traits to the study, e.g., to test if weather response for yellow spot in wheat is affected by the change of architectural traits, and the related change in microclimate.

More modelling work including sensitivity analysis will need to be done before being able to use the model to deliver decision support for managing plant pathogens by growers and industry, driven by climatic conditions, the overall aim of the study.

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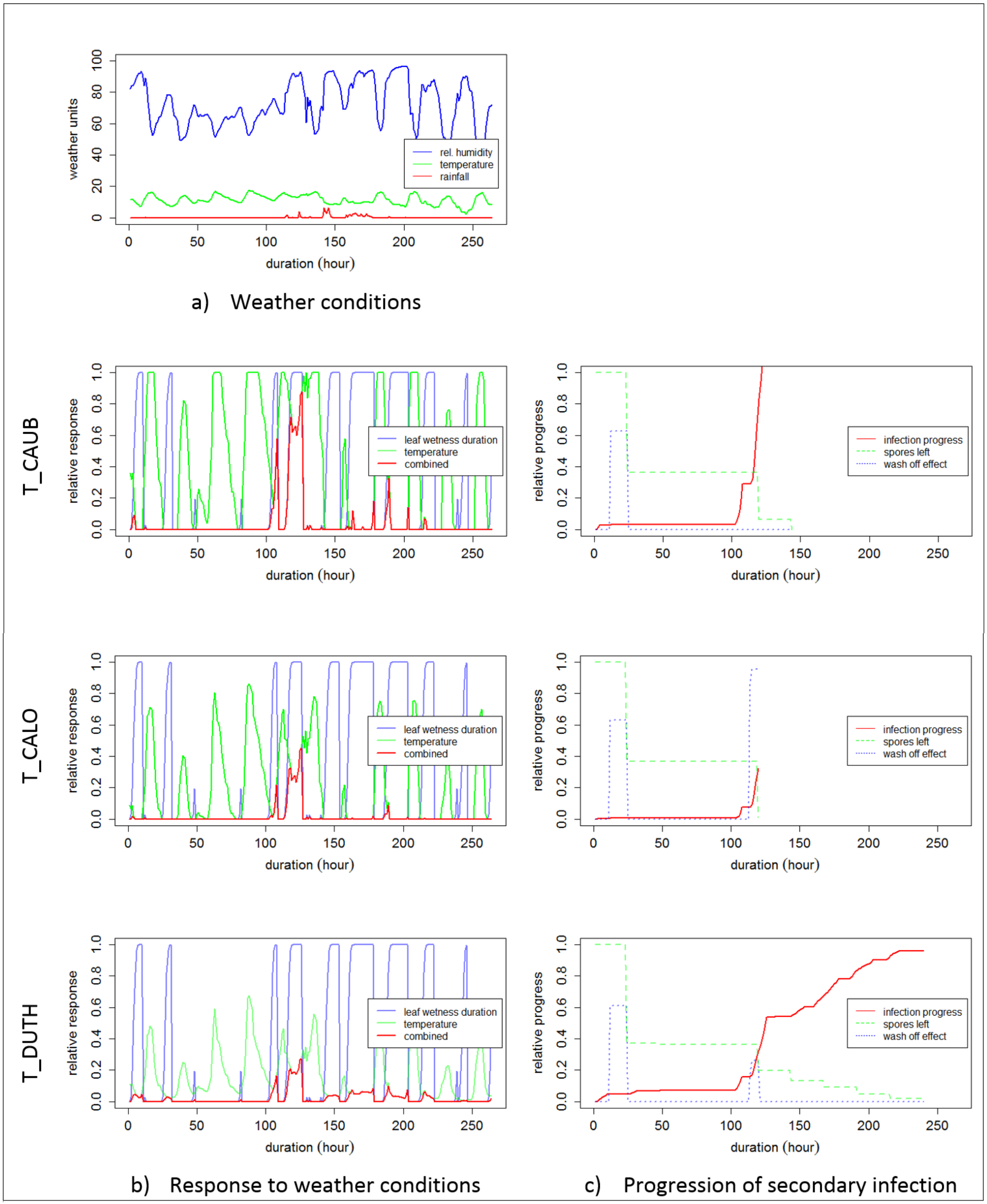
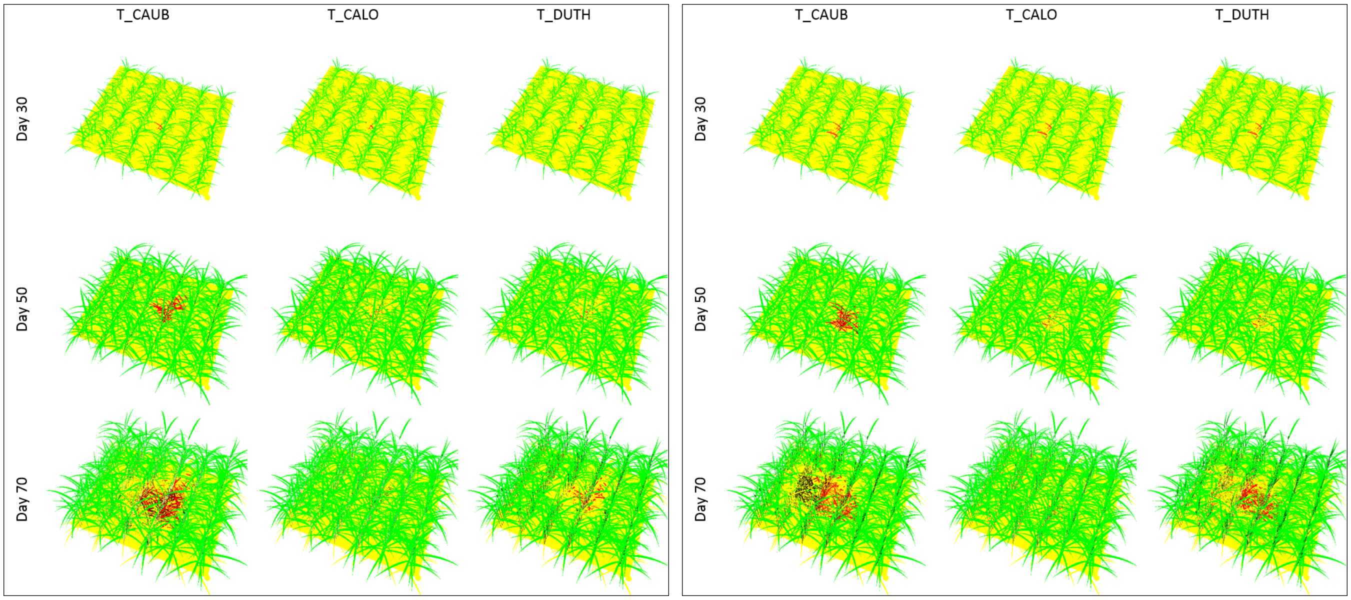


Fig. 8. Example of the progress of infection resulting from deposited spores after the simulated dispersal event on day 42 (03/07/2016), for three temperature response models and fixed RH_EXT, with the scale of one hour. a) Hourly weather data after day 42: relative humidity (blue), temperature (green) and rainfall (red); b) Relative response for leaf wetness duration (blue), temperature (green) and their combination (red). Value of 1 means optimal conditions; c) The progression of infection is calculated for all spores (conidia), released and deposited at the same time step. Difference between the progression at two consecutive days is used to calculate the probability of spores that caused new infection. The proportion of spores that caused new infection will be subtracted from the "pool" of released spores (green, updated after each day). In addition, weather conditions are evaluated and the loss of spores, e.g., by wash off (blue), is calculated, and subtracted from the pool. When the relative value of infection progression (red) is equal or greater one, the infection is considered as successful. Calculation of infection progression will stop if the pool is empty.



(a) Spore release rate 40% per day

(b) Spore release rate 100% per day

Fig. 9. Simulation output for three temperature response models, with fixed RH_EXT model, tracing the progression of disease from a single source lesion located in the middle of the canopy with 6x60 plants (red: leaf segments bearing lesions that became sporulating; yellow: leaf segments with latent lesions, or due to senescence; black: latent lesions).

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